(c) You are a bug sitting in one corner of a cubic room. You wish to walk (no flying) to the extreme opposite corner (the one farthest from you). Describe the shortest path that you can walk (draw diagram) and find out its length. [4]

Q.5) Attempt any two questions out of the following:

(a) Consider an example where four jobs (J1, J2, J3, and J4) need to be executed by four workers (W1, W2, W3, and W4), one job per worker. The matrix below shows the cost of assigning a certain worker to a certain job. The objective is to minimize the total cost of the assignment. Find out the optimal assignment by using Hungarian algorithm. [4]

	J1	*J2*	*J3*	*J4*
W1	82	83	69	92
W2	77	37	49	92
W3	11	69	5	86
W4	8	9	98	23

(b) Four points A(1,1) ,B(2,1),C(4,3) and D(5,4) are lying on a 2-dimensional X-Y plane. Group them into 2 clusters by using K-mean clustering algorithm. Show all the intermediate centroids and clusters in each iteration. [4]

(c)Term Frequency (TF) and Term Frequency-Inverted Document Frequency (TF-IDF) are the two methods in NLP to figure out the similarity between documents. Derive the expression to calculate TF and TF-IDF along with salient features of each one of them. [4]

Total No. of Pages: 04 Roll No.....

VI SEMESTER **B.TECH. [IT]**

END-SEMESTER EXAMINATION **(May - 2019)**

IT 306 ARTIFICIAL INTELLIGENCE & EXPERT SYSTEMS

Time: 3 Hours **Max. Marks: 40**

Q.1) Answer all the following questions:

(a) Discuss the experimental set-up and findings of Turing-test and Chinese room argument. [1.5]

(b) Draw AND-OR graph to solve the multiplication problem of 4 matrices($A_1A_2A_3A_4$). [1.5]

(c)How the local maximum, Plateau and Ridge problem in the state space of Hill Climbing can be solved? [1.5]

(d)Differentiate shallow and deep pruning in game trees with example. [1.5]

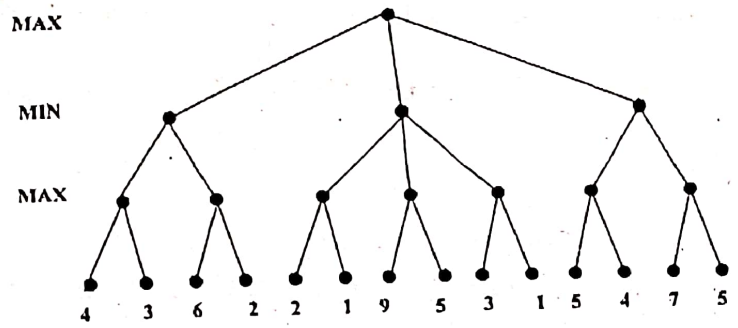
(e)How crossover and mutation can be performed to generate new solutions from the existing ones in Genetic Algorithm? [2]

Q.2) Attempt any two questions out of the following:

(a) You are given a set of scales and 12 marbles. The scales are of the old balance variety. That is, a small dish hangs from each end of a rod that is balanced in the middle. The device enables you to conclude either that the contents of the dishes weigh the same or that the dish that falls lower has heavier contents than the other. The 12 marbles appear to be identical. In fact, 11 of them are identical, and one is of a different weight. Your task is to identify the unusual marble and discard it. You are allowed to use the scales three times if you wish, but no more. Note that the unusual marble may be heavier or lighter than the others. You are asked to both identify it and determine whether it is heavy or light. [4]

(b)Apply alpha-beta pruning in the following tree . [4]

-36/-



(c) There are 100 light bulbs lined up in a row in a long room. Each bulb has its own switch and is currently switched off. The room has an entry door and an exit door. There are 100 people lined up outside the entry door. Each bulb is numbered consecutively from 1 to 100. So is each person. Person No. 1 enters the room, switches on every bulb, and exits. Person No. 2 enters and flips the switch on every second bulb (turning off bulbs 2, 4, 6...). Person No. 3 enters and flips the switch on every third bulb (changing the state on bulbs 3, 6, 9...). This continues until all 100 people have passed through the room. What is the final state of bulb No. 64? And how many of the light bulbs are illuminated after the 100th person has passed through the room? [4]

Q.3) Attempt <u>any two questions</u> out of the following:

(a) Following are the list of axioms in clause form:

1. man(Marcus)
2. Pompeian(Marcus)
3. born(Marcus, 40)
4. $\neg$man(x_1) V mortal(x_1)
5. $\neg$Pompeian(x_2) V died(x_2, 79)
6. erupted(volcano,79)
7. $\neg$mortal(x_3) V $\neg$born(x_3, t_1) V $\neg$gt($t_2 - t_1$, 150) V dead(x_3, t_2)
8. now=2008
9. (a) $\neg$alive(x_4, t_3) V $\neg$dead($x_4 - t_3$)
9. (b) dead(x_5, t_4) V alive(x_5, t_4)
10. $\neg$dead(x_6, t_5) V $\neg$gt(t_6, t_5) V dead(x_6, t_6)

Prove that Marcus is not alive now: $\neg$*alive(Marcus, now)* [4]

Draw resolution graph for the above proof.

(b) P and Q are two propositions. Which of the following logical expressions are equivalent? Justify your answer. [4]

I. $P \vee \sim Q$

II. $\sim(\sim P \wedge Q)$

III. $(P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge \sim Q)$

IV. $(P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge Q)$

(c) Write a PROLOG program to calculate the factorial of a number. [4]

Q.4) Attempt <u>any two questions</u> out of the following:

(a) Differentiate supervised, unsupervised and reinforcement learning methods by taking suitable examples. [4]

(b) Find the weights (till *Epoch 1*) using perceptron network for the following ANDNOT function when all the inputs are presented only one time. Use bipolar inputs and targets as shown below: [4]

x_1	x_2	t
1	1	-1
1	-1	1
-1	1	-1
-1	-1	-1